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Process and Device Simulation of 130 nm Generation CMOS Device Featuring 60 nm Transistors Operating at 1.4 V

Abstract

This TCAD Sentaurus™ simulation project provides a standard template for 130 nm generation MOSFET process simulation, device simulation, and electrical parameter extraction. The project is designed to serve as a template for TCAD engineers who want to explore the performance of 130 nm node CMOS devices.

Sentaurus Process simulates the MOSFETs with a physical gate length of 60 nm for the NMOS and PMOS devices, a 1.5 nm physical gate oxide, a 320 nm poly-to-poly pitch, and cobalt silicide as the device contact. The project setup features shallow trench isolation, retrograde wells, shallow abrupt source/drain extensions, and a halo and deep source/drain implantation process. The parameters and models for the process simulation are set by the Advanced Calibration models in Sentaurus Process. The simulated process flow and the device architecture are similar to the 130 nm generation CMOS device presented by Intel [1][2].

The electrical characteristics of the NMOS and PMOS devices are simulated using the Advanced Calibration set of models and parameters in Sentaurus Device for basic 130 nm node silicon MOSFETs (without any mechanical stress). The simulated electrical characteristics are similar to those reported in [1][2].

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Introduction

As the size of CMOS technology nodes decreases, shallow trench isolation (STI) becomes a crucial process aspect for achieving higher packing density and performance due to smaller channel-width encroachment, planar topography, and smaller junction-edge capacitance [2][3]. Similarly, retrograde wells, shallow abrupt source/drain extensions, deep source/drain implantations, and the salicidation process need improvement to allow for high-volume manufacturing with high-performance transistors after scaling [3].

In 2000–2001, Intel developed 130 nm generation CMOS technology with high-performance transistors that can operate in the range from 0.7 V to 1.4 V. Intel introduced a new process flow that starts with P^-/P^+ epitaxial silicon wafers, followed by the formation of STI. N-wells and p-wells are formed with deep phosphorus and shallow arsenic implantations, and boron implantations, respectively. A 450 nm deep trench isolation is formed to provide good intrawell and interwell isolation. High well doping is also used to control short-channel effects and to raise both the threshold voltage and the saturation drive current [1][2].

Sentaurus Process performs the process simulations of a generic 130 nm CMOS device using Advanced Calibration models for silicon CMOS, following the process recipe as presented in [1][2]. The process flow starts with the formation of a 450 nm deep STI, followed by the formation of a diffused twin-well. Finally, the halo and source/drain (S/D) implantations, and the thermal steps are optimized to achieve the required device geometry, p-n junction depth, and electrical characteristics as reported. Sentaurus Device simulates the electrical characteristics of the NMOS and PMOS devices using the Advanced Calibration models for a 130 nm CMOS technology node.

It is assumed that users are familiar with the Sentaurus tool suite, in particular, Sentaurus Workbench, Sentaurus Process, Sentaurus Device, and Sentaurus Visual, as well as the Synopsys IC WorkBench Edit/View Plus (ICWBEV Plus) tool. For an introduction and tutorials on specific tools, refer to the TCAD Sentaurus Tutorial.

General Simulation Setup

The simulation is organized as a Sentaurus Workbench project, where the tool flow is designed. Sentaurus Process executes several process steps to create the NMOS and PMOS devices, to adjust the meshing, and to place the contacts for device simulation. Sentaurus Device performs three separate device simulations, and Sentaurus Visual extracts all of the important electrical parameters.

Sentaurus Process

The tool sequence for this project starts with Sentaurus Process, which generates the MOSFETs. Sentaurus Process uses the following Sentaurus Workbench parameters:

- `Domain= NMOS | PMOS`: Defines whether the process flow for a 2D NMOS or PMOS device is required. The corresponding domains are selected from the layout.
- `Lgate` [µm]: Determines the physical gate length. It is used to stretch the layout and to parameterize the mesh definition. The gate length (drawn) of the target device is 0.06 µm (NMOS and PMOS).
- The subsequent parameters `Isolation`, `Poly`, `Ext`, `SD`, `Silicide`, and `Contact` are logical parameters that split the process flow into separate groups of simulation steps that are executed individually. This approach allows easy access to the intermediate structures for verification and allows you to study the process variability simulations for different process conditions without resimulating the previous processing steps.
- `DeviceMesh`: This is a logical split where all the necessary steps for the contact placement and the remeshing for device simulation are performed.

Sentaurus Device and Sentaurus Visual

Sentaurus Device performs the subthreshold characteristics (I_d–V_{gs}) simulations at a low-drain bias and high-drain bias and a family of output characteristics (I_d–V_{ds}) curves for two different gate biases. The bias conditions are controlled using Sentaurus Workbench parameters. The project extracts relevant electrical parameters as Sentaurus Workbench variables using a Sentaurus Visual script. The variables are:

- `Vtgm` [V]: Applied gate voltage where the transconductance is maximum.
- `VtiLin`, `VtiSat` [V]: Constant-current threshold voltages are extracted from the subthreshold characteristics.
- `IdLin`, `IdSat`, `Ioff` [A/µm]: On-state drain current from the linear and saturation regions, and the drain leakage current.
- `SSsat`, `SSlin` [mV/decade]: Subthreshold voltage swing for high-drain and low-drain biases.
- `gmSat`, `gmLin` [S/µm]: Maximum transconductance for high-drain and low-drain biases.
- `Ron` [Ω/cm^2]: On-state channel resistance.

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Tool-Specific Setups

Sentaurus Workbench and ICWBEV Plus

The Sentaurus Workbench split setup allows you to modify the process flow with minimal effort. All tools – Sentaurus Process, Sentaurus Device, and Sentaurus Visual – are invoked within the same Sentaurus Workbench session. The complete Sentaurus Workbench tool setup is shown in Figure 1.

		--	
SPROCESS	Domain	NMOS	PMOS
	Lgate	0.060	0.060
	Isolation	1	1
	Poly	1	1
	Ext	1	1
	SD	1	1
	Silicide	1	1
	Contact	1	1
	DeviceMesh	1	1
SDEVICE		--	--
	Vdd	1.4	1.4
	IdVg	1	1
SVISUAL		--	--
SDEVICE		--	--
	Vg1	0.7	-0.7
	Vg2	1.4	-1.4
	IdVd	1	1
SVISUAL		--	--

Figure 1 Sentaurus Workbench tool setup for 130 nm CMOS project

TCAD Sentaurus features an interface to the layout editor ICWBEV Plus, which allows for the viewing and editing of layouts in most industry-standard formats including GDSII and OASIS®. A parameterized layout is defined inside ICWBEV Plus. For details about its operation and capabilities, refer to the ICWBEV Plus documentation [4].

A generic CMOS cell layout is created in ICWBEV Plus following the reported design rules (`PolyPitch`, `IsoPitch`) for a 130 nm technology node [1][2]. Three layout-related files are present in the project directory:

- `CMOS.gds` – the GDSII layout file
- `CMOS_mkp.mac` – the markup file
- `CMOS130_lyt.mac` – the TCAD layout file

The markup file is created by users before the simulation and holds TCAD-specific information: the simulation domain, the stretch lines, and the auxiliary layers. The information is saved as an ICWBEV Plus macro file with the name `CMOS_mkp.mac`. This file can be loaded easily in the tool for further modifications if required.

Figure 2 shows the 130 nm CMOS cell layout as viewed in the user interface of ICWBEV Plus. In addition to the different layers (shaded polygons), you can see other objects specific to the current simulation. The white lines marked `NMOS` and `PMOS` represent the 2D simulation domains of the corresponding transistors. The white lines named `NMOS_L` and `PMOS_L` are stretch lines used to parameterize the device gate length. You can activate the 2D simulations for the NMOS and PMOS devices by switching to the appropriate simulation domain.

After the markup file is created, it is used as a source of information to generate the TCAD layout file, which contains a truncated version of the original layout with all of

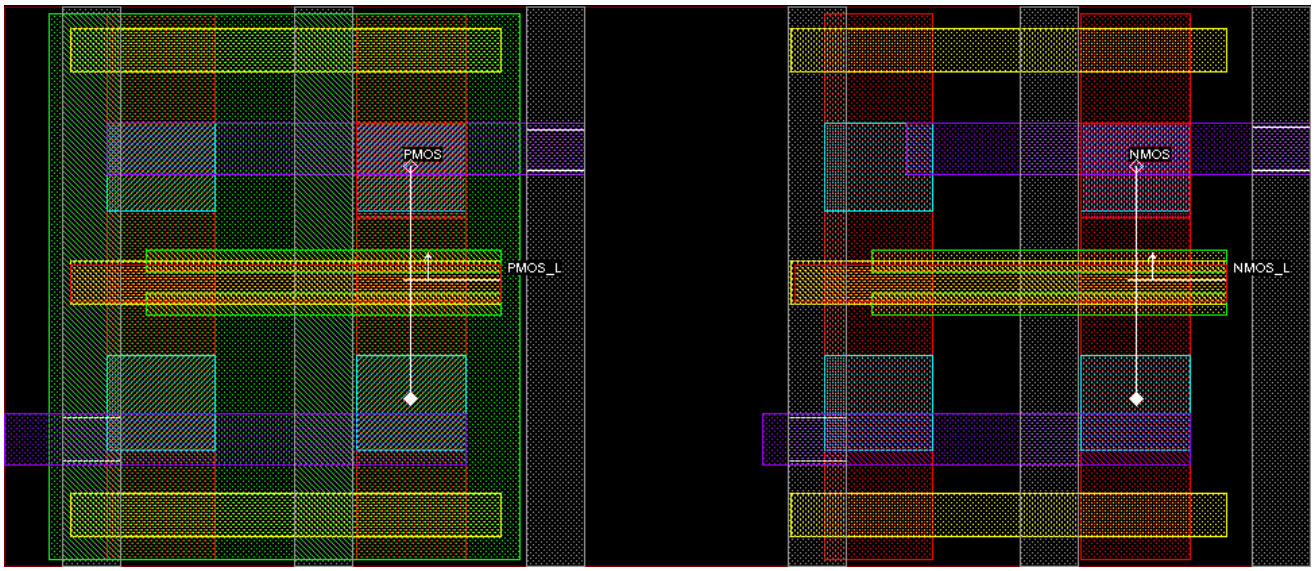

Figure 2 CMOS cell layout showing (*left*) PMOS markups and (*right*) NMOS markups

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the markup details. Then, this file is used as an interface to the process simulation:

```
icwb filename= "CMOS130_lyt.mac" scale= 1e-3
```

The simulation domain is declared as:

```
icwb domain= $Domain
```

where the `Domain` variables used in this project are `NMOS` and `PMOS`.

In Sentaurus Process, the `stretch` option is used in the following command:

```
icwb stretch name= "${Type}_L" \
  value= @<(Lgate-0.06)>@
```

Here, `Lgate` is the Sentaurus Workbench parameter. In this way, you can parameterize the simulation domain, the meshing setup, and the corresponding layout dimensions.

The bounding boxes of the simulation domain are extracted from the parameterized layout with these commands:

```
fset Ymin [icwb bbox left]
fset Ymax [icwb bbox right]
fset Ymid [expr ($Ymax-$Ymin)/2.0]
```

These coordinates are used later with the `line` commands to create the initial 2D simulation boundary.

Sentaurus Process

Sentaurus Process performs the process simulation part of the project. The following key characteristics of the process flow have been chosen for the 130 nm CMOS technology node [1]:

- Define the substrate with a (100) p-type epi-wafer and initial doping of 10^{15} cm^{-3}
- STI (0.45 µm silicon trench etching with sidewall slope of ~87°)
- Well and V_{th} adjustment implantations
- Patterning of 1.5 nm gate oxide and polysilicon gate
- Multi-step pocket/halo implantation, spacer formation, and S/D extension implantation
- Highly doped S/D formation (HDD) and rapid thermal anneal at 1000°C
- Cobalt silicidation for contacts

In this project, the process simulation uses the Advanced Calibration set of models and parameters by specifying:

```
AdvancedCalibration
```

To use the unified coordinate system in Sentaurus Process, specify the following command before the `line` commands:

```
math coord.ucs
```

Structure Formation and Meshing

The meshes are created by Sentaurus Mesh, which is the meshing engine of Sentaurus Process. Sentaurus Mesh and the MGOALS module control the bulk refinement and interface refinements. You can use the following command to redefine and update the meshing strategy:

```
grid set.min.normal.size= 0.005/$fp \
  set.normal.growth.ratio.2d= 2.0 \
  set.max.points= 10000000 \
  set.max.neighbor.ratio= 1e6
```

The initial simulation boundary is selected using `line` commands:

```
line x location= -0.1
line x location= 0.0 tag= top
line x location= 0.1
line x location= $Tsub tag= bottom
line y location= $Ymin spacing= 100.0 tag= left
line y location= $Ymid spacing= 100.0 tag= right

region Silicon substrate xlo= top xhi= bottom \
  ylo= left yhi= right
```

Sentaurus Process automatically generates the mesh as required using Sentaurus Mesh. Mesh generation starts with a bisection algorithm and, finally, the mesh elements are created using a modified Delaunay algorithm. In two dimensions, the entire structure is remeshed after each geometric operation.

Refinement Box for Meshing

The ICWBEV Plus–TCAD Sentaurus interface provides added flexibility to define refinement boxes and additional masks in this project. All types of positive and negative masks can be created in ICWBEV Plus using the following command:

```
icwb.create.all.masks
```

The masks are required to parameterize the device geometry and mesh. The refinements for the V_{th} adjustment and S/D implantations can be specified immediately before the corresponding implantation steps, to optimize computational resources. For example, in this simulation, the `Vt` and `HALO` refinement boxes are introduced before the implantation steps:

```
refinebox name= Vt mask= Active_p \
  extend= 0.015 extrusion.min= -0.001 \
  extrusion.max= 1.5 xrefine= "0.005 0.02/$fp" \
  yrefine= "$PolyPitch/(25.0*$fp)" \
  zrefine= "$IsoPitch/(20.0*$fp)"
```

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```
refinebox name= HALO mask= Active_p \
  extend= 0.01 extrusion.min= -0.05 \
  extrusion.max= 0.15 xrefine= "0.004/$fp" \
  yrefine= "$PolyPitch/(100.0*$fp)" \
  zrefine= "$IsoPitch/(60.0*$fp)"
```

Defining the STI Region and Polysilicon Gate in Sentaurus Process

Initially, 10 nm pad oxide and 100 nm nitride are deposited followed by photoresist deposition with the active area mask. Then, the STI region is created by trapezoidal etching:

```
etch material= {Silicon} type= trapezoidal \
  rate= $STI_depth time= 1.0 angle= $STI_tilt
```

The polysilicon gate is patterned using the following commands:

```
deposit material= {Poly} type= isotropic \
  thickness= $Tpoly
photo mask= POLY_n thickness= 0.2
etch material= {Poly} type= anisotropic \
  rate= $Tpoly+$gox time= 1.0
etch material= {Poly} type= isotropic \
  rate= {0.005} time= 1.0
deposit material= {Poly} type= isotropic \
  rate= {0.005} time= 1.0 selective.materials= {Poly}
```

The silicon recess etch near the polygate is formed by trapezoidal etching with a 45° angle and predefined rate:

```
etch material= {Oxide} type= anisotropic \
  rate= $gox time= 1.0
etch material= {Silicon} type= trapezoidal \
  rate= $SiRecess time= 1.0 angle= 45.0
```

The spacer in the structure is created by isotropic deposition and anisotropic etching of nitride with the same rate:

```
deposit material= {Nitride} type= isotropic \
  rate= {1.0} time= $Lsp
etch material= {Nitride Oxide} type= anisotropic \
  rate= {1.5*$Lsp 1.5*$Lsp} time= 2.0
```

Refer to the project *Sentaurus Technology Template: CMOS Processing* [5] for information about all other processing steps such as implantation, activation anneal, and silicidation.

Figure 3 shows the device structure with `NetActive` doping profiles, oxynitride cap layer, and silicide and tungsten as the contact `VIA` for the NMOS and PMOS transistors.

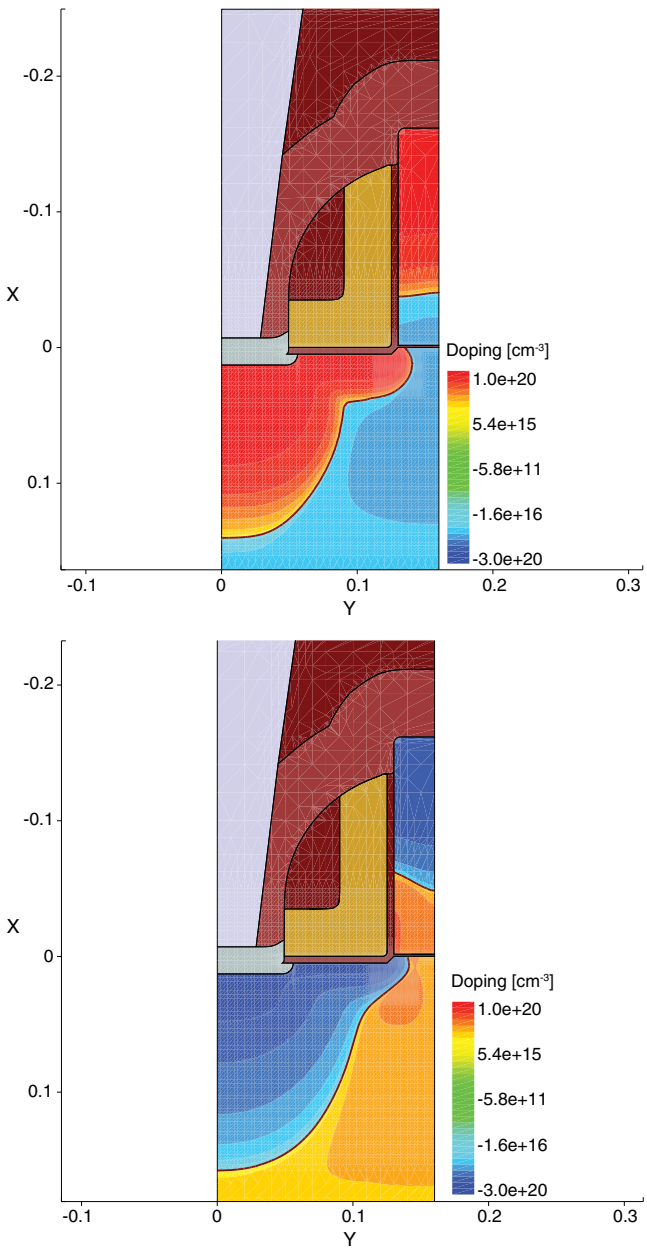

Figure 3 Simulated doping distribution for 60 nm gate length for (*top*) NMOS device and (*bottom*) PMOS device

Preparing the Structure for Device Simulation

At the end of the process simulation flow, Sentaurus Process prepares the structures for the subsequent device simulations. In the final split `DeviceMesh`, all the necessary steps for the final meshing and layout-driven contact placement are performed.

The meshing requirements for process and device simulations are different. For process simulations, it is convenient either to use a fixed meshing strategy or to adapt the mesh to the gradients of individual dopant profiles. For device simulations, these profiles are not of particular

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interest, but the profiles of the p-n junctions and junction resolution are very important. It is more useful to refine the net active doping concentration and to remove an unwanted portion of the substrate before the device simulation. Moreover, the channel, the gate oxide, and the channel–oxide interface must be resolved with a much finer mesh for device simulation. For this meshing, first, all of the previously defined refinement boxes are removed:

```
refinebox clear
refinebox !keep.lines
line clear
```

Second, Sentaurus Mesh is configured to create high-quality mesh elements that are best suited for device simulations:

```
grid Adaptive set.Delaunay.type= boxmethod \
  set.max.points= 1000000 \
  set.max.neighbor.ratio= 1e6 \
  set.min.normal.size= 0.01/$fd \
  set.normal.growth.ratio.2d= 4.0
```

Third, a new field-based adaptive mesh is created with the existing structure and doping information using Sentaurus Mesh inside Sentaurus Process. The adaptive and fixed refinement strategies are described by the boxwise refinement specification of Sentaurus Process. The refinement boxes to resolve the channel, the S/D region, and the interface regions are defined as:

```
refinebox name= Device_Active mask= Active_p \
  extend= 0.01 extrusion.min= -$Tpoly \
  extrusion.max= 0.3 refine.fields= {NetActive} \
  def.max.asinhdiff= 1.0 \
  refine.max.edge= "0.01/$fd $PolyPitch/(20*$fd) \
  $IsoPitch/(16.0*$fd)" \
  refine.min.edge= "0.0025/$fd $PolyPitch/(80*$fd) \
  $IsoPitch/(64.0*$fd)" \
  materials= {Silicon PolySilicon} adaptive

refinebox name= Device_Ext mask= Mesh_EXT_p \
  extend= 0.01 extrusion.min= -0.01 \
  extrusion.max= 0.04 xrefine= 0.002/$fd \
  yrefine= 0.002/$fd zrefine= $IsoPitch/(20.0*$fd) \
  materials= {Silicon}

refinebox name= Device_IF mask= Active_p \
  extend= 0.005 extrusion.min= -0.01 \
  extrusion.max= $Divot-0.002 \
  min.normal.size= 0.0002/$fd \
  normal.growth.ratio= 1.2 \
  interface.mat.pairs= {Silicon Oxide PolySilicon \
  Oxide}
```

Finally, the layout-driven contact placement command, `icwb.contact.mask`, is used to create contacts for the subsequent device simulations. For example, the commands for placing the source and gate contacts on the MOSFETs are:

```
icwb.contact.mask layer.name= "SOURCE" \
  name= source point CobaltSilicide x= $SilicideX \
  replace

icwb.contact.mask layer.name= "GATE" name= gate \
  box PolySilicon xlo=-$Tpoly-0.05 xhi=-$Tpoly+0.05
```

Figure 4 shows the final NMOS and PMOS device structures with field-based adaptive mesh and contacts used for device simulation.

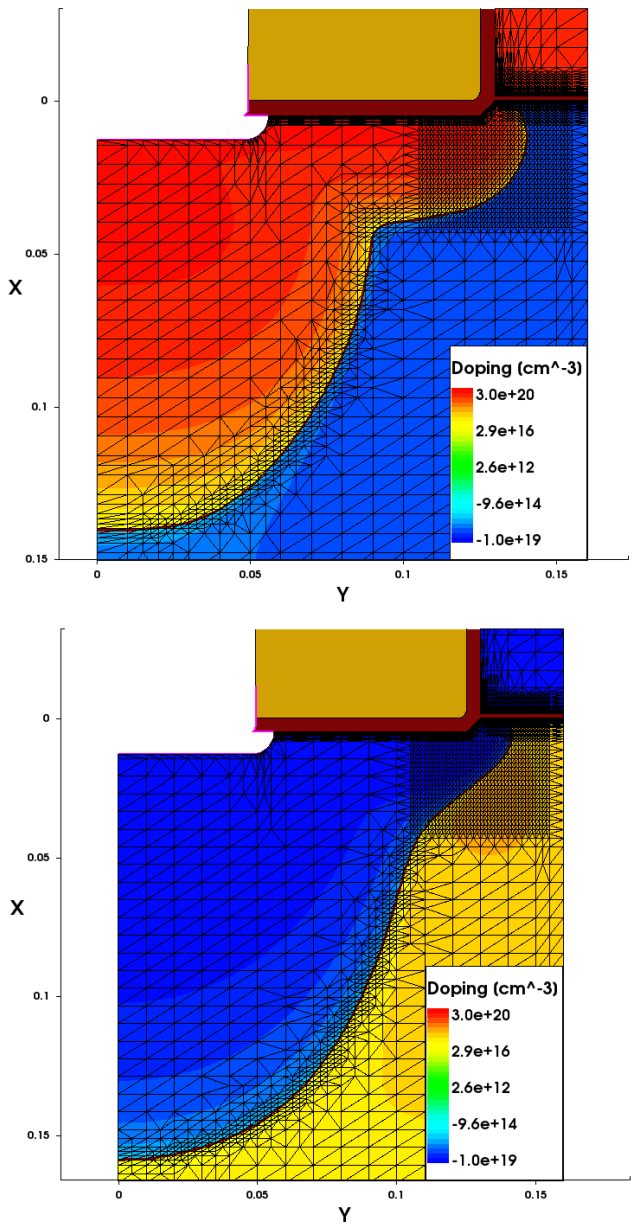

Figure 4 Final (*top*) NMOS and (*bottom*) PMOS device structures with contact and mesh generated by Sentaurus Process

Device Simulation Using Sentaurus Device

Sentaurus Device simulates the drain current as a function of the gate voltage at a low-drain bias $(I_d–V_{gs})_{lin}$ and a high-drain bias $(I_d–V_{gs})_{sat}$ as well as the output characteristics, that is, the drain current curves as a function of the drain voltage $(I_d–V_{ds})$ for two different values of the gate bias. The initial voltages at all four contacts of the device are defined in the `Electrode` section of the command file:

```
Electrode {
  { Name="source" Voltage= 0.0 }
```

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```
    { Name="drain" Voltage= 0.0 }
    { Name="gate" Voltage= 0.0 }
    { Name="substrate" Voltage= 0.0 }
  }
```

Sentaurus Device performs the device simulations using the Advanced Calibration set of models and parameters for silicon MOSFETs without any mechanical stress for 130 nm CMOS technology node. To use the Advanced Calibration of Sentaurus Device, you can look up the parameter files and the command files for specific device types and technology requirements as described in the *Advanced Calibration for Device Simulation User Guide* [6].

For all types of electrical simulation, the important models used in this project are:

- High-field saturation with the gradient quasi-Fermi potential driving force model
- Density gradient quantization model
- Inversion and accumulation layer mobility model
- Shockley–Read–Hall (SRH) recombination model

The mobility models for the NMOS device are specified in the `Physics` section of the Sentaurus Device command file:

```
Physics {
  ...
  Mobility(
    Enormal (IALMob(AutoOrientation))
    HighFieldSaturation
  )
  EffectiveIntrinsicDensity(OldSlotboom)
  Recombination(SRH)
}
```

Figure 5 and Figure 6 show the I_d–V_{gs} and I_d–V_{ds} characteristics, respectively, for the NMOS and PMOS devices at low and high terminal biases. The simulated electrical characteristics are similar to the experimental results reported in [1].

A leakage current of 56 nA/µm (for the NMOS device) and 42 nA/µm (for the PMOS device) is extracted from the simulation. The saturation currents are in the range of 1.1 mA/µm for NMOS and 0.55 mA/µm for PMOS at a supply voltage of 1.4 V.

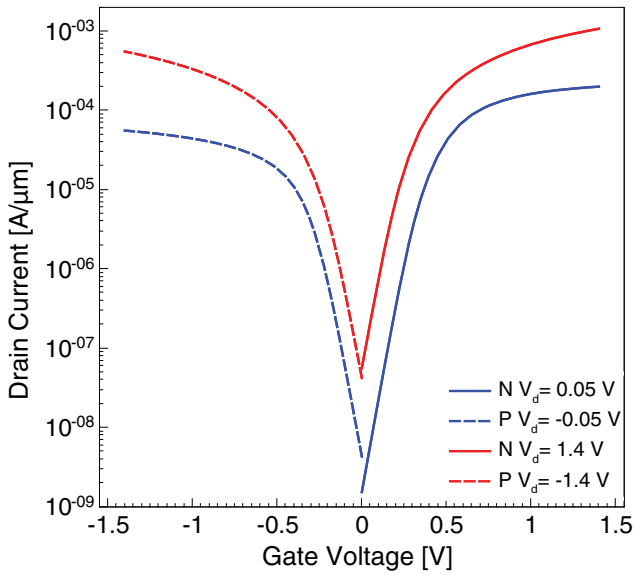

Figure 5 Simulated I_d–V_{gs} characteristics with drawn gate length of 60 nm; drain current as a function of gate voltage for (*left*) PMOS device and (*right*) NMOS device

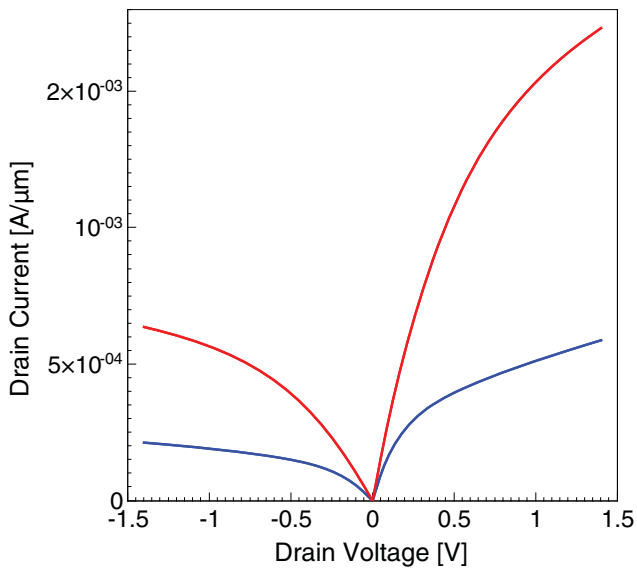

Figure 6 Simulated I_d–V_{ds} characteristics with drawn gate length of 60 nm; drain current as a function of drain voltage for (*left*) PMOS device and (*right*) NMOS device for a gate bias at 0.7 V (blue) and 1.4 V (red)

Figure 7 on page 9 shows the roll-off characteristics of the linear threshold voltage and the maximum transconductance voltage of the NMOS device. An optimal halo and well implantation recipe gives excellent roll-off characteristics for both devices. Similarly, S/D extension engineering is also performed to achieve a higher I_{on}–I_{off} ratio, as reported [1].

NOTE The roll-off characteristics simulations required different physical gate lengths (`Lgate`) and are not included in the project to reduce the project size, but they can be performed easily using the existing setup.

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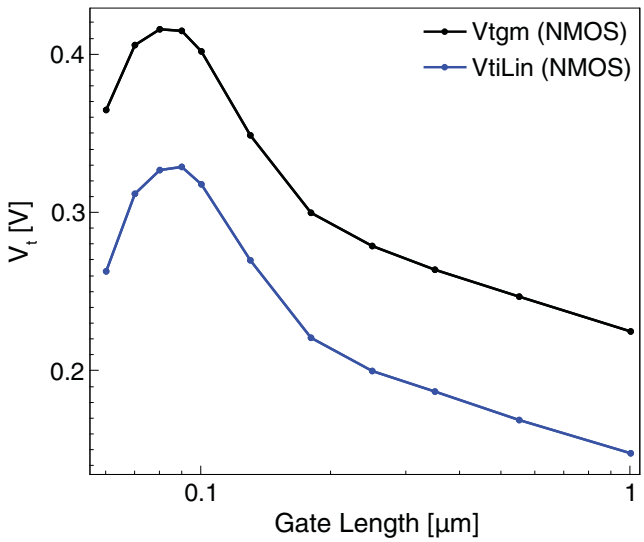

Figure 7 Threshold voltage as a function of gate length (roll-off) for NMOS device

Extraction and Visualization Using Sentaurus Visual

Each instance of Sentaurus Device in the Sentaurus Workbench tool flow is followed by an instance of Sentaurus Visual, which plots the corresponding I_d–V_{gs} and I_d–V_{ds} curves, and extracts the relevant electrical parameters. The extractions are performed using the internal libraries of Sentaurus Visual.

The following procedures of the extraction library are used in the project:

- `ext::ExtractVtgm` extracts the peak gm threshold voltage.
- `ext::ExtractVti` extracts the constant-current threshold voltage.
- `ext::ExtractIoff` extracts the drain leakage current.
- `ext::ExtractExtremum` extracts the drain on-current.
- `ext::ExtractSS` extracts the subthreshold slope.
- `ext::ExtractGm` extracts the transconductance.
- `ext::ExtractRdiff` extracts the channel on-state resistance.

Refer to the *Sentaurus™ Visual User Guide* and the "Extraction and Visualization Using Sentaurus Visual" section of the project *Sentaurus Technology Template: CMOS Characterization* [7] for more information about the extraction strategies.

References

[1] S. Thompson *et al.*, "An Enhanced 130 nm Generation Logic Technology Featuring 60 nm Transistors Optimized for High Performance and Low Power at 0.7 - 1.4 V," in *IEDM Technical Digest*, Washington, DC, USA, pp. 257–260, December 2001.

[2] S. Tyagi *et al.*, "A 130 nm Generation Logic Technology Featuring 70nm Transistors, Dual Vt Transistors and 6 layers of Cu Interconnects," in *IEDM Technical Digest*, San Francisco, CA, USA, pp. 567–570, December 2000.

[3] T. Ghani *et al.*, "Scaling Challenges and Device Design Requirements for High Performance Sub-50 nm Gate Length Planar CMOS Transistors," in *Symposium on VLSI Technology*, Honolulu, HI, USA, pp. 174–175, June 2000.

[4] *IC WorkBench Edit/View Plus User Guide*, Version N-2017.09-SP2, Mountain View, California: Synopsys, Inc., 2018.

[5] *Sentaurus Technology Template: CMOS Processing*, available from TCAD Sentaurus Version O-2018.06 installation, go to `Applications_Library/CMOS/CMOS_Processing`.

[6] *Advanced Calibration for Device Simulation User Guide*, Version O-2018.06, Mountain View, California: Synopsys, Inc., 2018.

[7] *Sentaurus Technology Template: CMOS Characterization*, available from TCAD Sentaurus Version O-2018.06 installation, go to `Applications_Library/CMOS/CMOS_Characterization`.

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